

Hybrid Vision Transformer and Mix Transformer Ensemble for Multiclass Brain Tumor Classification Using MRI Images

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Abstract—The classification of brain tumors utilizing Magnetic Resonance Imaging (MRI) represents an important aspect of computer-aided diagnosis, having direct significance for patient outlook and treatment planning. Conventional manual radiological assessment is time-consuming and often prone to inconsistencies, thereby creating a necessity for the creation of dependable automated methodologies. This work brings forward a hybrid deep learning approach, which combines the Vision Transformer (ViT) and Mix Transformer (MiT) architectures within a validation-optimized ensemble for multiclass brain tumor classification. [1], [2]. The MRI scans are rated into four categories, namely, glioma, meningioma, pituitary tumor, and no tumor by the system. Transfer learning is used on both models where the backbone unfreezing occurs in progressions, and is then trained with cosine learning-rate schedules that are warmed up and trained with Automatic Mixed Precision (AMP). The ensemble logit weighting is trained in the setting of a stratified validation set, where the weight coefficient of ViT is 0.6 and the weight coefficient of MiT is 1 0.6, and the test-time augmentation (TTA) is used to enhance the prediction robustness. Attention maps are used to facilitate the explainability. The proposed ensemble is tested where the public brain MRI dataset of 7,200 images with equal class distribution in 4 categories is used, the proposed ensemble achieves a test accuracy of 99.01% and a macro-averaged F1-score of 0.99 than all individual models [5].

Index Terms—Brain tumor classification, Vision Transformer, Mix Transformer, SegFormer, Ensemble learning, Transfer learning, MRI analysis, test-time augmentation, Explainability.

I. INTRODUCTION

Brain tumors are among the most dangerous neurological illnesses, which is defined by abnormal cell growth in the brain or near its place [6]. Effective treatment planning, surgical intervention and survival of a patient depend on early and accurate diagnosis. Magnetic Resonance Imaging (MRI) is widely preferred for brain examinations mainly due to its high soft tissue contrast imaging, and it also showcases much higher spatial resolution as compared to other imaging techniques.

Therefore, it continues to be the most important modality in the examination of brain tumors.

In spite of these benefits, radiologists still require a considerable amount of time to perform manual analysis of MRI scans, and the method is also subjective. The discrepancies in clinical experience, the fatigue, and ambiguity of interpretation cause inconsistencies in diagnoses. The computer-aided diagnosis (CAD) systems have become potential supplements to expert evaluation. Recent systematic reviews validate the growing adoption of machine learning (ML) and deep learning (DL) techniques in implementing brain tumor classification, with all cited studies depicting consistent improvements in accuracy compared to the conventional methodologies [6].

Initial computer-aided diagnosis (CAD) methods were based on manual feature engineering including texture descriptors, histogram-based features, such as support vector machines (SVMs) and k-nearest neighbors (KNN). Even though the traditional machine learning performs well in controlled environments, they do not perform well in cases that exhibit noise in the medical image scans, and show limited ability to generalize to unseen datasets. [4].

Convolutional neural networks (CNNs) made a significant contribution and made it possible to learn hierarchical features automatically. Transfer learning with architectures such as VGG, Inception, and ResNet were shown to have high tumor classification accuracy with the help of transfer learning [3], [7], [8], [11]. ResNet50, specifically, has been progressively hypothesized as a solid foundation on typical brain MRI benchmarks [3], [7], [8], [11].

However, CNNs are inherently local in nature, which constrains their ability to grasp long-range spatial information, which is essential to abnormally shaped tumors. Transformer architectures have been converted into a successful use in computer vision. ViT architecture captures long range dependencies by simply applying multi-head self-attention mech-

anisms over images that have been segmented into patches [1]. ViT-based models have demonstrated great performance and enhanced robustness in the analysis of brain tumors, in comparison to solely convolutional baselines [10], [11].

The Mix Transformer (MiT) was proposed as an encoder architecture that is the backbone of SegFormer offering hierarchical multi-scale feature representations, which is why it is suitable in structured visual classification tasks when local and global context are needed [2]. This paper is inspired by the complementary characteristics of ViT and MiT and will introduce a logit level ensemble of the two architectures to classify multiclass brain tumors. The framework includes progressive transfer learning, optimizes ensemble weights based on validation performance, test-time augmentation (TTA) using attention map visualization and horizontal flipping. Experiments convey that the proposed hybrid ensemble achieves elevated accuracy, strong class-wise performance, and improved interpretability.

Main contributions: A four-class brain tumor classification ensemble of ViT and MiT, optimized by the weighting at the logit level and validation, utilizing Magnetic Resonance Imaging (MRI). A progressive backbone training strategy based on cosine learning-rate scheduling with warmup and AMP on an NVIDIA T4. Empirical assessment of 99.01% accuracy of the test and macro F1-score of 0.99 with a balanced dataset of the population [5]. Qualitative analysis in terms of focus on clinically relevant tumor regions using attention map visualization.

II. RELATED WORK

A. Classical Machine Learning Solutions.

Traditional ML-based brain tumor classification systems rely on manually engineered features extracted from MRI scans, subsequently using shallow classifiers. Types of representative features are gray-level co-occurrence matrices, wavelet coefficients and shape or morphology descriptors. These are then passed to SVM, KNN or ensemble procedures [4]. A more recent systematic review, which found that such pipelines could perform reasonably on controlled datasets, found that they perform poorly at being robust and generalizing compared to end-to-end deep models [6].

B. CNN-Based Transfer Learning.

By transfer learning deep CNN features, Deepak and Ameer were able to classify MRI brain tumors, and this study showed that ImageNet-pretrained networks can be successfully transferred to a medical imaging task with minimum labelling data [3]. Bibi et al. and Younis et al. also demonstrated that high baseline accuracy can be achieved by fine-tuning ResNet50 on brain MRI, which frequently exceeds 95% on standard datasets, [7], [8]. Disci et al. conducted a detailed review of several pre-trained deep models, lightening their weaknesses and strengths [11]. In spite of these advances, CNNs are limited to local receptive fields and are not able to capture global context.

C. Vision Transformers in Brain Tumor Analysis.

Dosovitskiy et al. proposed ViT, which reformulates image

classification as a sequence modeling task using fixed size image patches, and uses global self-attention to add long-range interactions [1]. Tariq et al. suggest a hybrid ViT and EfficientNetV2 in the context of brain tumors and achieve more than 98% accuracy of multiclass brain tumor classification [10]. Disci et al. also found that ViT-based models are able to do better in standard benchmarks as compared to purely convolutional architectures [11].

D. Mix Transformer and Hierarchies.

SegFormer proposed the MiT encoder that synthesizes hierarchical feature maps across different resolutions and efficient attention with spatial reduction to cut on computational cost [2]. MiT features both the lower level of texture and the higher level of semantic data, which is why it is appealing when it comes to segmentation and classification of complex structures. Although MiT has found extensive applications in semantic segmentation, its application in brain tumor classification has not been relatively well studied.

E. Hybrid and Ensemble Methods.

The purpose of hybrid and ensemble approaches is to take advantage of the complementary strengths of different architectures. Hassan et al. presented a fuzzy-thresholding and deep learning hybrid system in brain tumor detection, and the results were competitive in the MRI dataset [9]. It was demonstrated that using ViT and EfficientNetV2 provides an increase in the performance that cannot be seen in the use of either of the two models alone (Tariq et al.) [10]. Simultaneously, Verbers et al. investigated the use of hyperspectral imaging in the detection of glioblastoma, where the next theme is the multi-modal fusion [12].

III. LITERATURE SURVEY

Brain tumors appear due to abnormal and uncontrolled cell growth within the central nervous system and can significantly affect neurological function and survival of the patient [6]. According to the grading system (Grades I–IV) by World Health Organization (WHO), the grade of the tumor reflects the level of malignancy and plays an important role in deciding the treatment strategies. In clinical practice, brain tumors are commonly categorized into four major classes: Glioma, Meningioma, Pituitary tumor, and No Tumor [5]. Magnetic Resonance Imaging (MRI) is broadly used as the primary non-invasive imaging modality for brain tumor diagnosis because of its ability to produce excellent soft tissue contrast and in-depth visualization of brain structures.

Despite the effectiveness of magnetic resonance imaging, the accurate classification of brain tumors remains difficult due to heterogeneous tumor structures, overlapping radiological features in different tumor categories, dataset imbalance, and differences in expert interpretation. Classical machine learning techniques and preliminary deep learning models, especially Convolutional Neural Networks (CNNs) and transfer learning architectures such as ResNet, have shown promising performance in automated brain tumor detection and classification [3], [7], [8], [11]. However, CNN-based approaches primarily focus on extracting local spatial features, which can limit

their ability to effectively capture long-range dependencies and global feature correlations within medical images.

Transformer-based architectures have recently shown significant potential in medical image analysis by effectively modeling global correlations through transformer-based attention modeling [1], [2], [10], [11]. Vision Transformers (ViT) effectively capture global contextual features, while Mix Transformers (MiT) introduce hierarchical feature representations that integrate both local and global information [1], [2]. Leveraging the complementary advantages of these architectures, this work proposes a hybrid ensemble framework that integrates Vision Transformer (ViT) and Mix Transformer (MiT) models for brain tumor classification from MRI images across multiple classes. By integrating predictions from both models, the proposed ensemble aims to improve feature representation, improve classification reliability, higher diagnostic accuracy compared to standalone models.

IV. PROPOSED METHODOLOGY

The proposed analytical frameworks will follow the following five stages: Image-wise stratified splitting and data set preparation. Vision Transformer (ViT) and Mix Transformer (MiT) fine-tuning on individuals with the use of progressive transfer learning. Horizontal flip averaging where the test time is augmented at inference. Production of attention maps with the aim of achieving post-hoc interpretability. Let z_{ViT} and $z_{MiT} \in \mathbb{R}^C$ denote the output logit of ViT and MiT respectively for a classification that involves a classification task for C classes. The ensemble prediction can be formulated as follows:

$$z_{ens} = \alpha z_{ViT} + (1 - \alpha) z_{MiT} \quad (1)$$

Here α is representing a weighting parameter with the range(0,1), it determines the relative contribution of each model during the ensemble prediction. To avoid the undesirable effects of softmax saturation, logit-level fusion is used instead of probability-level fusion.

V. DATASET AND PREPROCESSING

A. Dataset and Splits

This study utilizes the publicly available dataset which consists of over 7200 images that are classified by the model into four main categories; glioma, meningioma, pituitary and no tumor [5]. The pictures are assigned pre-established training and testing folders, where the classes are not skewed. The dataset structure is Training: 1,400 (glioma, meningioma, pituitary and no tumour) images per class, thus 5,600 Testing: 400 images for each class, totaling 1,600 images.

C. Preprocessing and Augmentation

All the MRI scans are processed in a standard image size of 224×224 pixels, that matches the input requirement of the transformer model, this step is then followed by normalization and augmentation techniques which help to further improve generalization. Model specific normalization presents the distributions of pixel intensities to ImageNet pre-training statistics. Data augmentation includes the horizontal flipping

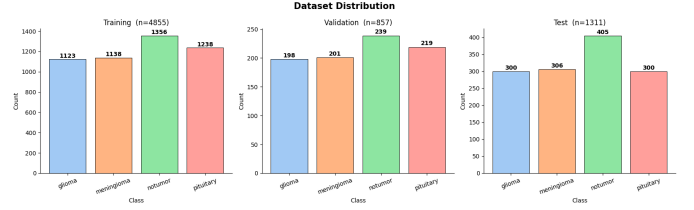


Fig. 1. Sample distribution across training and testing splits for the four MRI tumor categories class-wise.

and minor rotation within the range of $\pm 10^\circ$ which helps it to simulate variations while at the same time preserving the anatomical realism. (Fig. 1: Distribution of datasets class in terms of training and testing divisions)

VI. MODEL ARCHITECTURES

A. Vision Transformer (ViT)

ViT-Base configuration is implemented with a 16×16 patch size [1]. The 224×224 input image is divided into 196 discrete patches. these patches are then linearly embedded into 768 dimensions. They are fitted with learnable positional encodings. The resulting sequence operates through 12 layers of transformer encoders with each layer having 12 attention heads and a 3,072 feed-forward dimension [1]. The model performance is assessed and analysed that commonly adopt the classification metrics, including overall accuracy, recall, precision and the macro average F1 score.

B. Mix Transformer (MiT)

It uses the MiT-B2 setup that produces hierarchical feature maps at the stride levels of 4, 8, 16 and 32 in four encoding steps and channel sizes 64, 128, 320 and 512 [2]. The effective use of self-attention as well is integrated into each stage which is augmented with the spatial reduction ratios of 8, 4, 2 and 1. The extracted features of the final stage are, globally average-pooled and flowed to a linear classification head ($512 \rightarrow 4$ classes).

C. Ensemble Fusion

The logit-level fusion is used with $\alpha = 0.6$ being determined by grid search in validation set. Figure 2 below illustrates a proposed pipeline design for ViTMiT ensemble that will be developed and evaluated subsequently. (Fig.2: Proposed ViTMiT ensemble pipeline) This paper describes a proposed ViTMiT ensemble pipeline design that will be developed and tested in the future.

VII. TRAINING STRATEGY

A. Implementation Details

Tests were carried out on Google Colab with the usage of a single NVIDIA T4 card (16 GB). Both models were also trained in 50 epochs with a batch size of 32 and with AdamW (initial learning rate = 3×10^{-5} , weight decay = 0.1). The highest validation macro F1 scores were used to select checkpoints.

B. Transfer learning and Progressive unfreezing.

Both backbones have been loaded with ImageNet -pretrained weights. In the initial five epochs the backbone was frozen

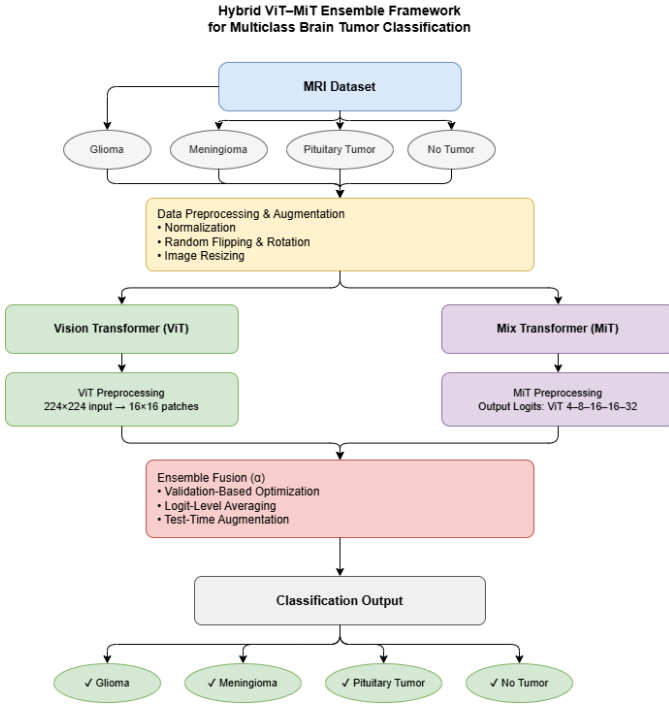


Fig. 2. Proposed hybrid ViT-MiT ensemble workflow for multiclass brain tumor classification, implementing a preprocessing pipeline, parallel transformer-based feature extraction streams, and logit-level fusion.

and the classification heads only updated. Between epochs 6 and 15, the transformer blocks were unfrozen. Starting with epoch 16, all the parameters were updated collectively. This gradual plan of action helps reduce catastrophic forgetting and establish a gradual adjustment of domains.

C. Learning Rate Schedule

A cosine decay schedule with a linear warm-up was used. The learning rate was then linearly increased from 1×10^{-5} to 3×10^{-4} in the first 500 warm-up steps, and then decreased linearly to 1×10^{-5} .

D. Regularisation and AMP

Label smoothing was used (cross-entropy loss, and the value of 0.1). Automatic Mixed Precision (AMP) helped minimize the use of memory and utilized T4 GPU Tensor Cores.

E. Test-Time Augmentation

All the test images were tested twice (one in the original orientation and the other with horizontal flipping). Averaging of ensemble logits was done before doing the softmax inference. (Fig. 3: loss and validation accuracy curves training)

VIII. EXPERIMENTAL FINDINGS AND REVIEW

Evaluation Protocol

The performance of the proposed hybrid Vision Transformer (ViT) and Mix Transformer (MiT) ensemble has been critically evaluated using a set of measures that are commonly applied in the field of medical image analysis and evaluation. The evaluation of the model is done on completely different and independent test dataset, this ensures that the results reported

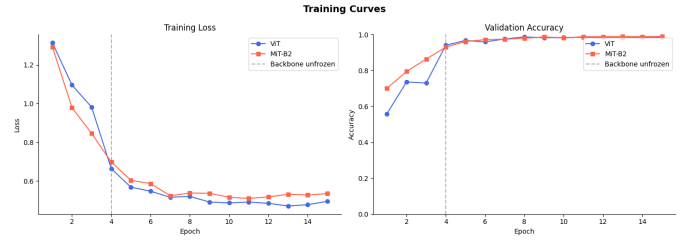


Fig. 3. The training loss and validation accuracy curves of both ViT and MiT models throughout 50 epochs along with progressive backbone unfreezing.

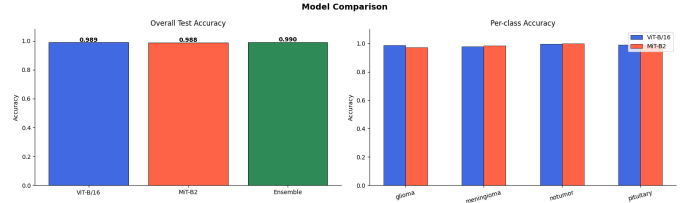


Fig. 4. Comparison of classification accuracy for ViT (alone), MiT (alone), and the proposed ViT-MiT ensemble in the held-out test set.

TABLE I
PER-CLASS CLASSIFICATION PERFORMANCE OF ViT-MiT ENSEMBLE (WITH TTA)

Class	Prec.	Rec.	F1	Supp.
Glioma	1.00	0.98	0.99	400
Meningioma	0.98	0.98	0.98	400
No Tumor	1.00	1.00	1.00	400
Pituitary	0.99	1.00	0.99	400
Macro Avg.	0.99	0.99	0.99	1600
Overall Accuracy	99.01%			1600

depict the model's generalisation capabilities. Evaluation was performed on an independent test set consisting of 1,600 MRI images. The dataset is balanced across four different categories, namely glioma, meningioma, pituitary tumor, and no tumor, with 400 images in each class. This balanced distribution ensured fair evaluation of the overall performance of the model in generalization. To ensure the model generalized to unseen data, the test data and the training and validation sets were completely separated.

Classification Performance

The proposed model that implements the ensemble framework has achieved an overall accuracy of 99.01%, this demonstrates the strong capability in distinguishing between the various categories of tumor. The model also attained a macro average F1 score of 0.99, which indicates balanced predictive performance across all classes of tumor. A slight reduction in precision and recall is observed in the meningioma class. This difference can be explained by the fact that the morphological characteristics of meningioma and glioma tumors in MRI images are often visually similar and therefore difficult to distinguish. Slightly lower performance is observed in the classification of meningioma mainly because of its similarity between the glioma MRI scans that are observed visually.

Confusion Matrix Analysis

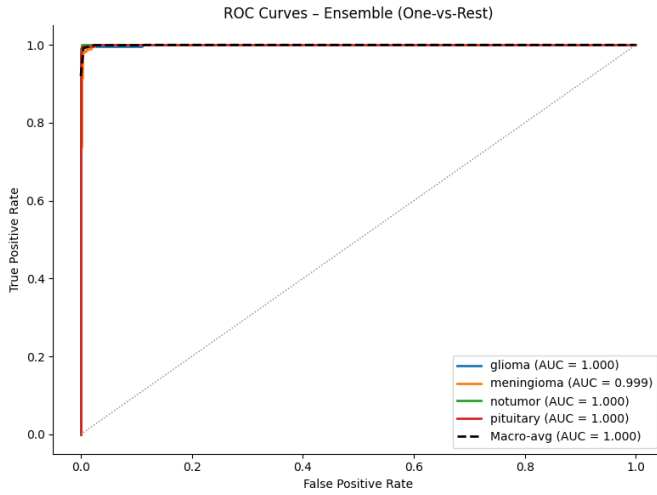


Fig. 5. One-versus-rest ROC curves for all the tumor classes along with macro-average AUC of the proposed ViT-MiT ensemble with Test-Time Augmentation.

The confusion matrix provides an in-depth view of the performance of the model in classifying brain tumors from the MRI scans. In a confusion matrix most of the predictive values are placed along the diagonal elements of the matrix, these elements represent the correctly classified samples. This tells us that the model is able to accurately determine the different brain tumor classes through the scans in the MRI dataset. Only a few instances appear in the off diagonal cells which represent the misclassifications. These errors are mainly observed in the tumors that are visually identical in MRI scans such as glioma and meningioma. To sum it all up the dominance of the diagonal elements of the confusion matrix depicts the robustness and reliability of the proposed classification model

ROC Curve Analysis

The Receiver Operating Characteristic (ROC) analysis is used to evaluate the ability of the model to correctly differentiate between the positive and negative instances across different classification thresholds. For each tumor category, the ROC curves show a strong separation between true positive and false positive. Values close to 1.0 for most tumor classes reflect excellent performance in the area under the curve (AUC), A higher AUC score suggests that the model has high sensitivity and high specificity, this enables the model to accurately identify and classify tumor cases while at the same time minimizing false detection.

Ablation Study

An ablation study has been conducted to analyze the individual contribution with the proposed framework of the different components. In this study, the performance of the Vision Transformer(ViT) and Mix Transformer(MiT) have been combined in an ensemble model and thoroughly compared. These standalone models were evaluated independently to understand the strengths individually, in terms of feature extraction, overall accuracy and classification. In the end the ensemble model that integrated both ViT and MiT was tested to determine

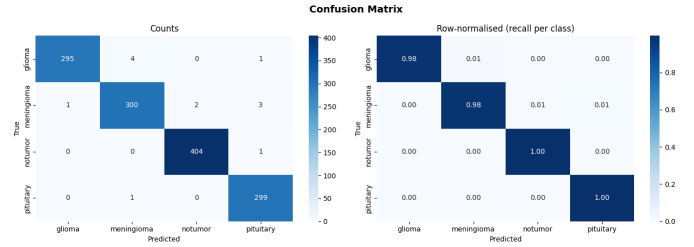


Fig. 6. Confusion matrix of the proposed ViT-MiT ensemble on the test set where the raw prediction counts are met on the left and the count of row-normalized recall values are met on the right.

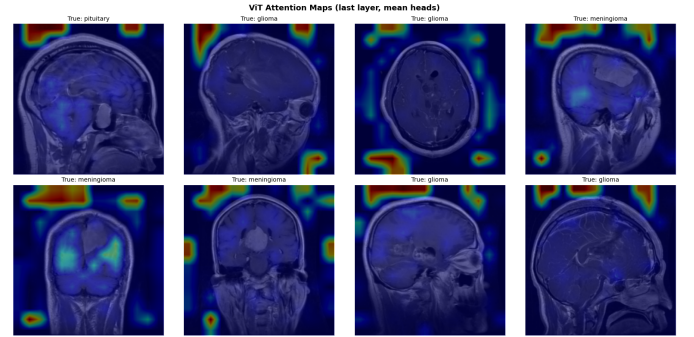


Fig. 7. Attention map overlays from the last ViT encoder layer (mean across attention heads) presented on sample MRIs.

whether the combined models would provide complementary capability leading to improved performance. The ensemble approach provides prominent classification accuracy and robustness.

Attention Map Visualization

Attention maps were derived from the transformer layers that provide visual insights into the regions of the MRI images. The idea behind these maps is that it highlights the areas that receive the highest attention weights, that means it indicates the parts of the image that have strong influence in the model's prediction. These highlighted areas are mainly considered as the boundaries of the tumor or abnormal tissue growth that are present within the uploaded brain scans. This alignment depicts that the model is learning the necessary medical information, rather than relying on background information, which is considered unnecessary. All in all the attention map not only improves the interpretability, but also increases the model's confidence in its decision making for brain tumor classification.

IX. DISCUSSION

ViT captures holistic spatial context effectively, whereas MiT is an efficient multi-scale tumor morphology encoder [1], [2]. The domain adaptation had to be stabilized with the progressive unfreezing as a necessity due to the relatively limited size of the data set. Meningioma is the most difficult type because it morphologically resembles glioma. The observed absolute gains (approximately 0.15 percentage points) point to the directional advantages but not to the statistically strong

TABLE II

ABLATION STUDY — INDEPENDENT CONTRIBUTIONS OF ENSEMBLE AND TTA

Configuration	Test Accuracy (%)
ViT alone (no TTA)	98.86
ViT alone (with TTA)	98.93
MiT alone (no TTA)	98.78
MiT alone (with TTA)	98.85
Equal-weight ensemble, $\alpha = 0.5$ (no TTA)	98.89
Optimized ensemble, $\alpha = 0.6$ (no TTA)	98.93
Optimized ensemble, $\alpha = 0.6$ (with TTA)	99.01

All configurations use the same test split and best-validation checkpoint.
Differences $< 0.25\%$ should be treated as indicative trends.

superiority, which confirms the need to analyze the data with multiple seeds. Application of this method in practice will face challenges confronted by scanner heterogeneity, volumetric data that are inherently 3-D, and multi-institutional variability. These dimensions should be clearly covered in future studies.

X. CONCLUSION

The proposed ensemble model that implements both ViT and MiT demonstrated highly effective performance in the task of recognizing brain tumors from MRI images. The global contextual learning capability of ViT combined with the multi-scale feature extraction ability of MiT allows the model to capture detailed local patterns as well as capturing broader structural relationships for the brain MRI scans [1], [2]. This combined learning approach helps differentiate between different types of tumors, while maintaining higher model test accuracy of 99.01%. The high test accuracy indicates strong reliability and stability in tumor classification. The model uses ImageNet-pretrained backbones, along with progressive layer unfreezing and cosine learning-rate scheduling. Training is conducted using Automatic Mixed Precision (AMP) on a T4 GPU. Validation-optimized logit fusion generated final predictions, which is further improved by test-time augmentation. Future research includes exploring ensembles with random initialization to achieve statistical strength and robustness, to extend the model to three dimensional MRI data, and seek clinical validation to test its real-world applicability.

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